

4.4 What are some of the key empirical limitations of the CAPM? ↕

The Limitations of the CAPM

Despite its widespread industry adoption, the CAPM has several serious limitations.

A quick way to see its limitations is to look at the OLS regression form of the CAPM from section 4.1:

$$R_s - R_F = \alpha + \beta(R_M - R_F) + \epsilon$$

If CAPM is the true model, one expects to estimate $\alpha = 0$ in the above regression. However, the estimated α is not just different from 0 but has also been found to vary with factor such as size (market capitalization), value (price-to-book ratio), and momentum (recent winners vs recent losers). This suggest that the CAPM, despite being useful, is far from a complete picture.

For a great introduction to the limitations of the CAPM, see the following article:

Fama and French (2004), "The Capital Asset Pricing Model: Theory and Evidence", *Journal of Economic Perspectives*, Vol. 18, No. 3, pp. 25-46. This article is available [here](https://learn.usc.edu.au/courses/35359/files/3049430?wrap=1) (<https://learn.usc.edu.au/courses/35359/files/3049430?wrap=1>).